Bracknell College Engineering	Student Name: Callum B Date: 10/11/2025
	Assessor Name: Jim Chuter Date: 10/11/2025

T Level Engineering 2025-27

Assessment Task Number: MA 1.

It is important that you are ready to complete this assessment activity. Talk to your assessor if you are not. In order to pass this assessment you must demonstrate your competence by completing the practical task without advice or guidance. Assessment will include observation of you performing the task and questions to determine your under-pinning knowledge. If you are observed working in an unsafe manner the assessment will be immediately cancelled.

This assessment requires that the following are completed:

1) Standard planning sheets: - Procedure. Tools and Equipment. Hazards and Controls. Engineering Drawing.

2) Practical task – Drill, Hole and Thread gauge.

Safety:

1. Always adopt safe working practices and procedures.
2. Use all personnel protective equipment supplied.
3. Follow all COSHH regulations according to the HASAWA.

Procedure:

1. Obtain all relevant documentation before starting work.
2. Collect and inspect all tools and equipment needed to complete the job.
3. Prepare material for work. Clean and debur material. Mark all dimensions according to diagrams.
4. Carry out all cutting, shaping and machining tasks.
5. If there are any problems seek advice from the lecturer, however attempt to have a solution to the problem before seeking advice.
6. Tidy away all tools and clean work area for good housekeeping reasons.

3) Inspection Sheet
4) Observation/Oral Question Sheet – Where appropriate
5) Written Questions

Engineering T Level
Planning Sheet/ Job Report

Name of Student: Callum Bartaby

Job Title: Drill Hole and Thread Gauge Task number: MA1

Procedure:

- Start by filing down any rough edges to make a piece of low-carbon steel square (you can verify this by using an engineer's square).
- Next, mark your left datum of 10mm using a digital height gauge.
- Mark a bottom datum line of 10mm.
- Next mark a datum line of 25mm along the left.
- Mark a 25 mm datum line along the bottom.
- Next mark a 35mm datum line along the left.
- Mark a 45mm datum line along the bottom.
- Mark a left datum line of 50mm.
- Mark a bottom datum line of 60mm.

- Mark a hole using a centre punch 10mm in from the vertical datum line either end of the gauge.

- Using the centre punch again, mark holes in the bottom 8 meeting points of the datum lines.

- Drill these marked holes 4mm in diameter using a pillar drill.

- Then, drill a 5mm hole in the second bottom pilot hole.

- Drill a 6.5mm hole in the third bottom pilot hole.

- Drill a 8.5mm hole in the far-right bottom pilot hole.

- Drill a 10.2mm hole in the top left pilot hole.

- Drill a 8.2mm hole in the second top pilot hole.

- Drill a 6.1mm hole in the third top pilot hole.

Procedure continued:

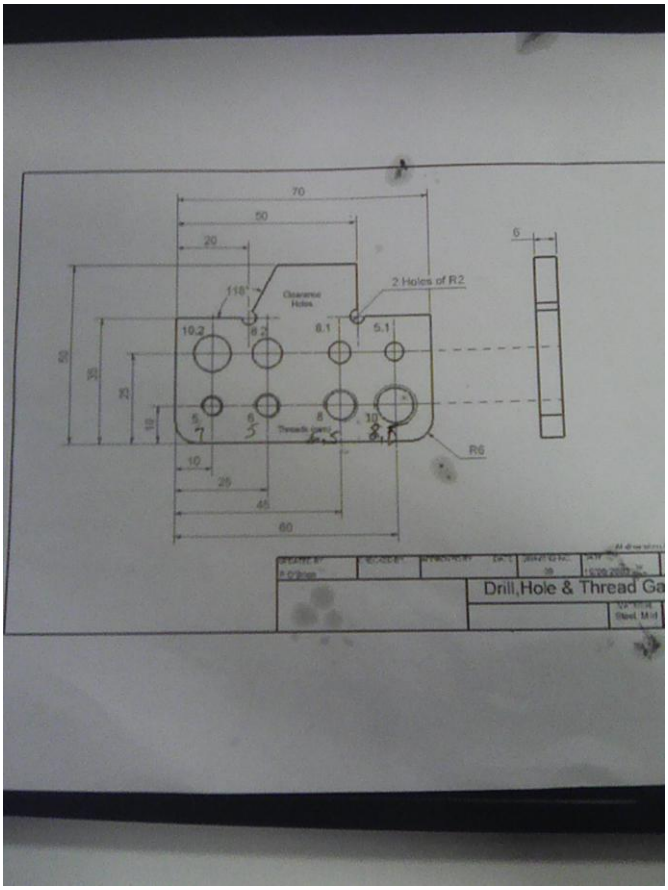
- Drill a 5.1 mm hole in the top right final hole.
- Next, draw a line 118^0 line diagonally left from the left uppermost hole 10mm in from the left datum, and mark with a centre punch.
- Similarly, from the right uppermost hole 10mm from the right datum, draw a line at exactly 90^0 vertically to the top of the gauge.
- Use a hacksaw to cut out both lines, leaving the top shape seen in the diagram in the front page.
- File down these cut edges to smoothen and straighten.
- Using a tap-drill conversion chart to ensure you get the right size tap, in order to input the correct size threads within each bottom hole.
- Round off the bottom corners with a file to a radius of R6.
- Finally, draw file both flat sides of the gauge to take off the top layer of carbon, giving the gauge a shiny finish.

Problems encountered/Assistance required:

Misfiled the cut off edges and removed my R2 holes.

Needed assistance rounding off the corners to R6 as I was unsure what I was doing.

Documentation, Tools and Equipment Requirement Sheet				
Documentation requirement:				
✓Engineering Drawing	☐Engineering Sketch	☐Procedure Sheet		
☐Other-Specify				
Work holding equipment requirement:				
☐3 Jaw Chuck	✓Bench Vice	☐Angle Plate	☐G-Clamp	
☐4 Jaw Chuck	☐Drill Press Vice	☐Plain Clamp	☐Collet Chuck	
☐Other-Specify				
Measuring and Marking out equipment requirement:				
✓Vernier Callipers	✓Engineer Square	✓Protractor	✓Radius Gauge	
☐Micrometer	☐Torque Wrench	☐Steel Rule	☐DTI	
☐Surface Table	☐Straight Edge	☐Template	☐Dividers	
☐Thread Gauge	☐Marking Blue	☐ Other-Specify		
Machine equipment requirement:				
☐Box and Pan Folder	☐Cornice Folder	☐Fly Press	☐Notcher	
☐Milling Machine	☐Bench Bender	✓Pillar Drill	☐Guillotine	
☐Pinch Roller	☐Rotary Saw	☐Swaging M/C	☐Lathe	
☐Other-Specify				
Tool requirement:				
✓Scriber	✓Centre Punch	☐Dividers	✓Hammer	☐Mallet
✓Hack-Saw	✓Files	✓Drills/Hole-saws	☐Deburrer	☐Sockets
☐Spanners	☐Snap Set	☐ Knurling Tool	☐Riveting Block	☐ Twist Pliers
☐Air Drill	☐ Forming Tool	☐Chamfer tool	☐Riveting Gun	☐Air Angle Drills
✓Tap	☐Wire Striper	☐Cutting Tool	✓Punch /Die	☐Reamer
☐Die	☐Allen Keys	☐Countersink Cage	☐Centre Drill	☐Parting off Tool
☐Other-Specify				


70
50
20
6
2 Holes of R2
118°
Clearance Holes
10.2
6.2
8.1
5.1
50
35
25
10
5
7
6
5
8
10
8.1
Threads pitch
R6
10
25
45
60
Drill,Hole & Thread Ga
Steel, Mild

Date: _____